T20 WORLD CUP 2022

SQL Data Analysis

Uncovering batting, bowling & match insights through structured query language (SQL)

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Data Tables

20
SQL Queries

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Concepts Used

Project Overview

This project performs an end-to-end SQL analysis of ICC T20 World Cup 2022 data, transforming raw cricket records into actionable insights across four interconnected tables.

`bating_summary`

Ball-by-ball batting: runs, balls, SR, 4s, 6s per match

`bowling_summary`

Bowling figures: wickets, economy rate, maiden overs

`match_summary`

Match outcomes, venues, dates, winners

`t20_players_2022`

Player profiles: team, role, batting style

MySQL 8.0+ | **Window Functions** | **JOINs** | **Date Transforms** | **Aggregations**

Schema & Data Preparation

Before running analysis, the schema was cleaned and dates were properly typed:

```sql
-- Rename columns for clarity
ALTER TABLE bowling_summary
RENAME COLUMN
economy TO economy_rate;

-- Parse date strings into DATE type
ALTER TABLE match_summary
ADD COLUMN Match_Date DATE;

UPDATE match_summary
SET Match_Date =
  STR_TO_DATE(
matchDate, '%b %d, %Y');
```

1. **Rename Columns**
   economy → economy_rate
   SR → strike_rate

2. **Add Date Column**
   Added Match_Date (DATE type) to match_summary

3. **Transform Dates**
   STR_TO_DATE() to parse 'Oct 16, 2022' format

4. **Disable Safe Mode**
   SET SQL_SAFE_UPDATES = 0 for bulk UPDATE

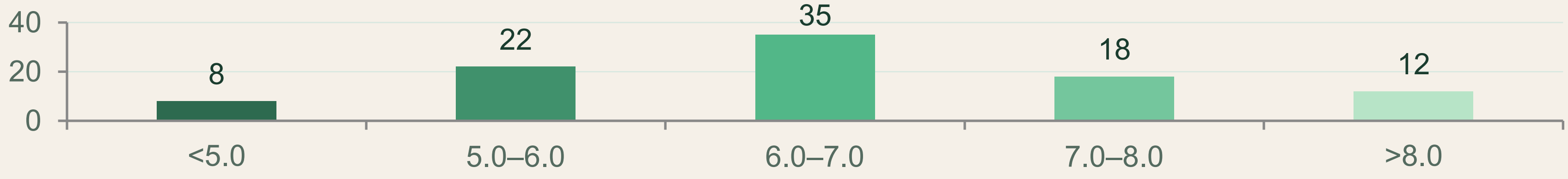
Batting Analysis — 8 Queries

Q3	**Highest Score**	*Max runs in a single match + match identifier*
Q9	**Player Averages**	*AVG(runs) per player, ordered by performance*
Q10	**Best Power Hitters**	*Top 2 by runs with fewest balls consumed*
Q11	**Sixes by Playing Role**	*SUM(6s) grouped by all-rounder / batsman / etc.*
Q12	**50–80 Scorers**	*Batsmen between 50 and 80 runs, sorted by match*
Q16	**SR > 200**	*Ultra-aggressive batsmen with strike rate > 200*
Q19	**Six-Hitters Ranking**	*DENSE_RANK() on total sixes, handles ties fairly*
Q6	**Left-Hand Batsmen**	*JOIN to filter by battingStyle = 'left hand bat'*

Bowling Analysis — 5 Queries

Q4	**Top Wicket Taker**	*Highest wickets in a single match + match context*
Q8	**Maiden Over Bowlers**	*Bowlers with maiden ≥ 1, ordered by maiden count*
Q15	**Economy Rate < 5**	*Most economical bowlers across the tournament*
Q18	**Avg Economy Rate**	*ROUND(AVG(economy_rate), 3) — tournament-wide stat*
Q20	**Wickets Ranking**	*DENSE_RANK() on total wickets across all matches*

Tournament Economy Overview



Match Insights — 5 Queries

Q2	**Team Win Count**	*COUNT(winner) per team, sorted by wins descending*
Q13	**Weekend Matches**	*DAYNAME(Match_Date) IN ('Saturday', 'Sunday')*
Q14	**Matches per Ground**	*COUNT grouped by ground to find busiest venues*
Q17	**Abandoned / No Result**	*Matches where winner = 'abandoned' or 'No result'*

Illustrative Win Share — Top Teams

Key SQL Concepts Demonstrated

JOINs

INNER JOIN between player profiles and performance tables using shared name key.

Window Functions

DENSE_RANK() OVER (ORDER BY ...) for ranking batsmen and bowlers without gaps.

Date Handling

STR_TO_DATE() parses text like 'Oct 16, 2022' into MySQL DATE type.

Aggregations

COUNT, SUM, AVG, ROUND used with GROUP BY for team & player-level stats.

Schema Alteration

ALTER TABLE to rename columns and add new typed columns to existing tables.

Filtering

WHERE with BETWEEN, IN, comparisons and DAYNAME() for date-based filtering.

Summary & Takeaways

- 20 SQL queries spanning batting, bowling, match, and player dimensions
- Applied window functions (DENSE_RANK) for proper tie-safe rankings
- Date transformation pipeline from raw text to typed DATE column
- INNER JOINs linking player profiles with live performance data
- Filtering by role, style, economy, strike rate, and match outcome

🏏 **20 Queries**

📋 **4 Tables**

⚡ **6 SQL Concepts**